

| Question |     |      |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |          |          |           |           |          |
|----------|-----|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|-----------|----------|
|          |     |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total     | Maths     | Prac     |
| 7        | (a) | (i)  |      | X-rays penetrate matter <b>or</b> muscle <b>or</b> tissue (1)<br>Denser matter not penetrated as much <b>or</b> absorbed by bone (1)<br>X-rays affect photo film / photo cells (1)                                                                                                                                                                                                                                        | 3               |          |          | 3         |           |          |
|          |     | (ii) | I    | Rearrange to get $v = \sqrt{\frac{2eV}{m}}$ (1)<br>$v = 7.95 \text{ [or } 8.0] \times 10^7 \text{ [m s}^{-1}]$ (1)                                                                                                                                                                                                                                                                                                        |                 | 2        |          | 2         | 2         |          |
|          |     |      | II   | $P = 18 \times 10^3 \times 12 \times 10^{-3} = 216 \text{ [W]}$ (1)<br>X-ray power = $0.005 \times 216 = 1.08 \text{ [W]}$ (1)                                                                                                                                                                                                                                                                                            |                 | 2        |          | 2         | 2         |          |
|          | (b) | (i)  |      | $I = \frac{I_0}{2} \rightarrow \frac{I_0}{2} = I_0 e^{-\mu x} \rightarrow \frac{1}{2} = e^{-\mu x} \text{ or } \frac{1}{2} = e^{-\mu x}$ (1)<br>Logs taken of both sides and convincing algebra (1)                                                                                                                                                                                                                       |                 | 2        |          | 2         | 2         |          |
|          |     | (ii) |      | $\mu = 0.257 \text{ or } 0.26$ (1)<br>Correct use of logs <b>or</b> $\ln 0.7 = \mu x$ (1)<br>$x = 1.39 \text{ or } 1.4 \text{ [cm]}$ (1)                                                                                                                                                                                                                                                                                  |                 | 3        |          | 3         | 3         |          |
|          | (c) |      |      | Ultrasound B-scan would be very effective [only] form of imaging to give moving images <b>or</b> can monitor blood flow (1)<br>Fluoroscopy also very effective with contrast medium can give a series of rapid images over time / view in real time (1)<br>MRI (would image soft tissue) no moving images (1)<br>X-rays poor at imaging soft tissue (1)<br>CT scans poor at soft tissue imaging [high radiation dose] (1) |                 |          | 5        | 5         |           |          |
|          | (d) |      | (i)  | Absorbed dose is the energy absorbed per kg of tissue (1)<br>Equivalent dose is the absorbed dose multiplied by a weighting factor due to the type of radiation used (1)                                                                                                                                                                                                                                                  | 2               |          |          | 2         |           |          |
|          |     |      | (ii) | $4 \times 0.12 = 0.48 \text{ mSv}$ <b>unit mark</b>                                                                                                                                                                                                                                                                                                                                                                       | 1               |          |          | 1         | 1         |          |
|          |     |      |      | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                                                                   | <b>6</b>        | <b>9</b> | <b>5</b> | <b>20</b> | <b>10</b> | <b>0</b> |